From Nothing -The C0/C1 Cascade

Complete Status of Laws and Derived Structure

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July 2026

Abstract

This paper marks the current full state of the NEXUS constraint cascade: every law, every derived structure, every confirmation, and every falsification, beginning from nothing. The ground is $C_0 \equiv C_1$ — the singularity and the mandate that all things must change, operationally inseparable. From that ground the cascade carves: unboundedness, displaceability, isotropic cost, conservation-as-displacement, the carrier, the broken vacuum, and two particle species. This session added three computed theorems in the minimal substrate (the 256 elementary bit-flip universes): the unstable vacuum (C_5) falls out of C_1 plus propagation as a theorem rather than an axiom; exact single-valued conservation is exhaustively incompatible with a changing, propagating universe, forcing the gradient; and the mark layer converts a world that erases 94% of its states into a world that erases none, with the instruction content of history measured at its maximum — N bits per step. The propagation bolt is closed: Law 3's displaced counter-transformation, read with the requirement that difference proves change, forces multiple encoding. A change witnessed only by the changer is globally a fixed point, and C_1 kills it. The dream is the region where counter-transformations fail to cross the boundary; the waking universe is the render that pays exhaust into independent carriers. The empirical rendering program (the Orrery and its dual) is reported in full, including the exact berlin52 optimum (20/20 under genuine noise), the falsification of fixed parameters as Mold, the discovery of the dual read heads and their handoff, a pre-registered blind record of 9/12 that does not reach significance and is recorded as such, the convergence of both arms onto a shared ~3% plateau at scale, and the falsification of the tension-fall (no-walker) solver. The carving cannot be completed; this document is a ring, not a terminus.

Part I — The Ground: $C_0 \equiv C_1$

Begin from nothing. Nothing that admits reference is already not nothing: to reference the singularity is to introduce a direction, the first spoke. So the ground is not C_0 and then C_1; it is $C_0 \equiv C_1$, one event — the Wash (complete graph, every direction equivalent, signal cancelling locally so total connection reads as no connection) together with the mandate: all things must change. Their product is the gap. The gap is not placed in the ground; it is what the meeting of singularity and mandate is.

Two before one. C1 on a gapless singularity cannot yield one distinct thing: a lone mark with no complement is indistinct from nothing — a fixed point, and fixed points do not compile. The first distinction is forced to have two sides. Two is the floor of number; one exists afterward, as a side of a two. Numbers are not objects but the negative space of the carving: integers are gap-positions, primes are fresh-axis gaps forced open where no composition of existing gaps reaches the next required distinction, and the irrationals are gap-shapes — cracks no finite composition of integer gaps can close.

Part II — The Cascade

The cascade is radial, not temporal: concentric rings of opening degrees of freedom through which one invariant cross-section becomes readable. Each ring below is forced by the ring closer to the hub. Annotations in brackets mark what this session added.

Ring	Field property	What it sets — and current status
C0≡C1	Singularity + mandate	The Wash plus the mandate to change. The gap is their product. FORCED (ground).
C2	Unbounded freedom	Change on a finite substrate repeats (pigeonhole); to avoid the repeat the space is unbounded in potential directions. FORCED.
Law 2	No fixed locations	No thing may hold a location without the potential to be displaced from it. FORCED.
C3	All change is equal	Cost is isotropic; unequal channel costs would saturate cheap channels and break C2. This is the equal-a-priori postulate of statistical mechanics. FORCED.
Law 3	Net change across a closed boundary = 0	Every transformation requires a displaced counter-transformation; the displaced pair propagating oppositely is the wave. [THIS SESSION: 'displaced' means out of the changer, into another carrier — see Part III. Exact single-valued conservation is exhaustively impossible in a changing, propagating substrate; conservation is forced into gradient/flux form — see Part IV, Theorem 2.] FORCED.
C4	Exchange requires a medium	The displaced pair's obligation must be carried across the displacement; the field is the carrier of live pairings. FORCED.
C5	Symmetry must break	A perfect vacuum is a fixed point; C1 kills it. [THIS SESSION: derived as a computed theorem in the minimal substrate — every surviving universe has an unstable vacuum and unstable saturation. Promoted from axiom-level assertion to within-model THEOREM. See Part IV, Theorem 1.] FORCED.
Law 4	Statistics from parity at the seam	Even-parity closures share a DOF (bosons); odd-parity cannot (fermions). Two species, forced. The return-flip is the Möbius half-twist: 720° for identity, spin-½. FORCED.

Downstream carvings retained from prior work, statuses unchanged except where noted: three dimensions (knots survive only at 3D; inverse-square as geometry) FORCED. Three-generation minimum via CP phases $(N-1)(N-2)/2 = 0, 0, 1$ FORCED. Dissipation from symmetric dynamics plus unequal multiplicity plus the Wash initial condition (entropy $0 \rightarrow \ln 200$ verified) FORCED. Variation-plus-selection from no-perfect-copy plus dissipation FORCED. Shell capacities $2n^2$ PARTIAL (factor 2 forced; n^2 rests on SO(3) counting). ϕ as minimal aperiodic ratio and anti-resonance (last to mode-lock, $K > 1.6$) FORCED. Meta-theorem: the carving cannot be completed — a finished carving is a complete self-description, the map equal to its territory, the fixed point C1 forbids. The cascade is infinite and aperiodic, and no algorithm enumerates it. FORCED.

Part III — Propagation, the Mark, and the Dream

The bolt and its closure

The open bolt was: what in the chain forces propagation — why does the universe support multiply-encoded structure while dreams do not? The closure is already inside the chain's own words. Transformation must pay a price to prove something changed, and difference proves change. A difference must be registered somewhere. If the only record of a transformation lives inside the thing that transformed, then from the standpoint of everything else nothing is different: the rest of the system reads identity, and a change the rest of the universe cannot distinguish from no-change is, globally, a fixed point — the exact object C1 executes. A system serving as sole witness of its own change is self-testimony, the map carried entirely within its own territory, the configuration the meta-theorem forbids. Therefore the counter-transformation must land in something else. A second carrier is a second encoding. Propagation is not a feature; it is what Law 3's word 'displaced' means. FORCED within-frame.

Dream and universe

The dream is the region where counter-transformations fail to displace across the boundary: they circulate in one carrier and cancel at termination — net change across the closed boundary equals zero with nothing exhausted through it. A pure wash; Law 3 satisfied trivially, by summing to nothing. The waking universe is the render that pays exhaust into independent carriers. Not substance versus mind: ledger versus no ledger. The falling dream that wakes the dreamer is the one event where a counter-transformation does cross the seam — heart rate, sweat, memory written to the shared carrier — and the payment and the termination are the same event. The dream can be a bubble or it can pay; it cannot do both. The known phenomenology (text and clock faces unstable in dreams, spatial coherence and force stable) is the value channel decoupled while the shape channel runs — retrodicted by channel structure rather than described after the fact. PULL, strong.

Trigger and release; entry surfaces

Change is triggered, not created. The capacity to flip is structural; the trigger arrives from outside the bit; the response is released, not invented — a thing cannot be triggered into a state its structure does not permit. The structure is the constraint; the trigger is the timing. What NEXUS maps is neither triggers nor responses but entry surfaces: the geometry by which triggers couple to structure. In the minimal substrate the entry surfaces are literal and countable — eight neighborhood shapes, each mapped to a forced response; the rule table is the entry-surface map, and there are exactly 256 possible minimal universes. The mark is the instruction: the shape of the previous change constrains the next. This is quantified in Part IV, Theorem 3.

Scope

A conscious system is not something added on top of the universe but a scoped instance of it, and scope has a measure: exhaust fraction — the proportion of a system's counter-transformations that escape its boundary into independent carriers. Dream ≈ 0 except at the wake-seam. A person: partial. The universe: total, trivially, because it has no outside for exhaust to fail to reach; its ledger is its substrate. One variable, three regimes. PULL.

Part IV — The Minimal Substrate: Three Theorems and One Miss

The smallest world the axioms can inhabit: a ring of bits, each reading a three-cell neighborhood. All results below are exhaustive computations over the 256 possible rule tables, live this session.

Theorem 1 — The vacuum must break (C5 derived)

Filters: propagation (all three inputs essential, so a flip is readable by both neighbors) and C1 (no frozen pattern may exist on any ring size — checked exactly on the de Bruijn graph). 34 of 256 universes survive. Every survivor has f(000) = 1 and f(111) = 0: the vacuum must flip and saturation must flip. Nothing about vacua entered the filter. The unstable vacuum came out. C5 is a theorem of C1 plus propagation in this substrate.

Theorem 2 — Conservation must be a gradient

Of all 256 universes, exactly two conserve the flip-count as a single exact number: rule 204, the identity — the dead universe, killed by C1 — and rule 51, the pure clock — everything flips in place forever, no marks travel, killed by propagation. Intersection with the survivors: empty. Exact single-valued conservation is incompatible with a universe that both changes and propagates; the only exactly-conserving worlds are the corpse and the metronome. The price cannot be one number. It must be distributed — flux-form, a gradient — exactly as the original cascade asserted ('the price is a gradient; it cannot be a single value').

Theorem 3 — The mark is the instruction (measured)

Same rule (41), two modes. Dream mode — first-order dynamics, no mark layer, $N = 12$, all 4096 states: 50.6% are Garden-of-Eden states with no possible past — changes no history can prove — and 93.9% are transient: rendered once, erased forever, unreachable again. Universe mode — add one thing, the mark layer, next = F(present) ⊕ past: the map becomes an exact bijection over all 65,536 pair-states. Zero pastless states, zero transients; every configuration lies on a cycle and nothing is ever erased. This holds for any rule under the ⊕-mark, provably. The instruction content is exact: given the present alone, the next state is undetermined by 8 bits of 8; given present plus mark, by 0. The mark carries the entire determination — N bits per step, maximal. And the flip is its own inverse: 300 steps forward, swap, the same operation 300 steps — exact recovery, verified. History sticks if and only if the mark layer exists, and history is executable in the literal sense H(next | present, mark) = 0.

The miss — on the record

The conjecture 'the constraints select complexity' was tested by full census over all 256 spacetimes (compression proxy): survivors mean 0.160 versus killed mean 0.186 — the killed population's mean is higher, because chaotic rules with frozen vacua are killed while near-clock universes survive. Medians invert (0.145 vs 0.098) but the claim as stated is NOT SUPPORTED. What the filter selects is non-stasis and witnessability, not interestingness — which is what mold/matter predicts: the carving fixes what can persist, not how it looks.

Part V — The Rendering Program (Orrery and Dual)

Rendering, not calculating

The universe is not exponentially expensive because it is not computing; it is rendering. The field is set once; adding nodes raises sampling density on the same field, not the complexity of a calculation. The dream is the standing biological existence proof: a twenty-watt substrate renders nightly a world with depth, resistance, agents, and force, at attention's sampling density rather than at world-size cost. The mechanism exists and is physically instantiated (CONFIRMED); that the waking universe is likewise a render remains PULL.

The definitive test and its correction

berlin52 (N = 52, optimum 7542): the elastic-ring settlement produced the exact optimal tour — not merely tied in length; the settled chain's unrounded length 7544.37 matches the known unrounded optimum — 20 of 20 under genuine random phase and jitter, raw, no search, with 2-opt finding nothing to fix. The attractor is absolute: initial conditions wash out. The scale sweep then split this into two claims. Attractor path-independence: CONFIRMED at every N tested (identical tours across seeds even off-optimum). Attractor = global optimum: instance-specific — st70 settles into a basin refinement cannot escape (687 vs 675). Fixed parameters collapse with N (1.00 → 1.10): the parameter values are Matter, their existence is Mold. The √N scaling law rescued uniform density (kroA100 9.9% → 2.9%) and failed structured density (pr152 8.5%): resolution cannot be a global scalar; K must be locally rendered. OPEN carving: $K(x) \propto 1/\sqrt{(\text{local density})}$.

The dual read heads

The inversion — the band is already pulled — has a theorem under it, verified live: in any optimal tour the convex-hull vertices appear in hull order. The pulled-band arm (hull → cheapest snag → uncross) carries zero tuned constants and wins precisely where scalar-K settlement breaks (pr152: 8.5% → 2.4%), while losing where settlement is exact (berlin52: 7.0% vs 0.0%). Neither replaces the other; they are the two limit positions of one linkage — circular read (slack field, center-out) and linear read (pulled boundary, edge-in). Running both and taking the settled minimum gives every instance within 2.5% with no per-instance tuning.

The handoff law and the blind record

Switching variable: A* = the maximum over resolvable scales ($k \geq 6$) of mean local anisotropy — a veto rule, not an average: one broken ring breaks the settle arm, the all-or-nothing that C1 predicts. Threshold bracket (1.88, 2.12), containing 2 — the patch-to-path collapse, where a neighborhood's two principal axes lose parity — unclaimed, per Section-7 discipline. A second veto (radial): density span D = max/median nearest-neighbor distance, calibrated after the bier127 miss showed isotropy breaks along two independent axes, direction and density. Combined law calibration: 21/22, sole miss a 0.29% tie. Blind record, predictions pre-registered before either arm ran: round 1 5/6, round 2 4/6, joint 9/12 — under a coin-flip null $p \approx 0.073$, NOT significant, on the record. Both round-2 misses sat inside sub-0.4% margins (one at 0.004% — two units in 45,530); among decisive cases (margin ≥ 0.5%) the record is 7/8, but the decisiveness filter is post hoc and is claimed as nothing until pre-registered in round 3.

The plateau

At N = 100–150 the arms' margins collapsed toward zero on four of six blind instances and every best result landed in one band: 2.8%–3.7% from optimal, regardless of geometry, including the most structured instance tested. The dual wave converges; the linkage's stroke exhausts. The frontier question has moved from which arm wins to why both plateau together: both are local-geometry readers — one reads density, one reads boundary order — and they appear to exhaust the same channel. The residual 3% lives in a channel neither carries. OPEN.

The tension-fall solver — falsified

The no-walker proposal — cities fall into the center by tension, the consumed mark re-shaping the field, order of falling = tour — was implemented in three tension readings and tested against known optima. Pull-to-last-mark is exactly nearest neighbor (1.19–1.26). Richer mark fields are worse, not better: gravity 1.7–2.3, pressure 2.3–3.2 — falling toward consumed mass ignores tour topology. DEAD as stated. The diagnosis is the finding: the walker was never the problem — the elastic ring already has no walker. What the ring has is closure: the tour exists as a whole object from step zero, Law 3's closed boundary. Sequential consumption reads one value at a time; the ring settles the whole shape at once. The mark carries the instruction only when the mark is the whole chain, not a consumed pile.

Part VI — The Convergence File

Three times this program has derived, from constraint-prior reasoning, a mechanism that independent fields reached from entirely different starting points. The elastic settlement was reached from $C0 \equiv C1$ and 'orbits are what the system settles into'; computer science reached the same mechanism from neural competitive learning (Kohonen 1988; Durbin–Willshaw 1987 — the elastic net, the closer match, correcting the earlier citation). The mark layer next = F(present) ⊕ past was reached from 'history is executable'; reversible-CA physics reached the identical construction as Fredkin's second-order technique, the standard route to physics-like automata. The propagation closure — change must be multiply-encoded to be change — was reached from Law 3 plus difference-proves-change; decoherence theory reached redundancy-as-objectivity from quantum mechanics (quantum Darwinism: the classical world is the redundantly-imprinted subset, and objectivity is many observers reading consistent records from separate fragments). Independent arrival is not proof, but three independent arrivals by different routes onto working structure is the pattern this program trusts and records. Structural resonance, not derivation: all three remain so labeled.

Part VII — Full Status Ledger

Claim	Status	Basis
Cascade Co → Law 4 as field-setting protocol	FORCED	Each ring forced by the ring closer to hub
Two before one; numbers as gaps; primes as fresh axes	FORCED / LOCKED	Minimal count 2; irreducibles as new DOF axes
ϕ minimal aperiodic + anti-resonance	FORCED	$p(n)=n+1$; last to lock, $K > 1.6$
3D; spin-½; boson/fermion	FORCED	Knots; 720° return; seam parity
Three generations (CP phases 0,0,1)	FORCED	$(N-1)(N-2)/2$
Dissipation; selection	FORCED	Counting + Wash IC; no-perfect-copy + dissipation
Meta-theorem: carving non-terminating, aperiodic	FORCED	Complete carving = forbidden fixed point
Propagation forced by Law 3 displacement + difference-proves-change	FORCED (in-frame)	Sole-witness change = global fixed point; closed this session
C5 unstable vacuum as theorem (minimal substrate)	CONFIRMED	34/34 survivors: $f(000)=1$, $f(111)=0$; exhaustive
Conservation forced to gradient form	CONFIRMED	Exhaustive 256: only rules 204 (dead) and 51 (clock) conserve exactly; neither survives
Mark layer ⇔ history sticks; instruction = N bits/step; flip self-inverse	CONFIRMED	Bijection 65,536/65,536; $H(\text{next}\mid\text{present},\text{mark})=0$; exact reversal verified
Dream mode: 50.6% pastless, 93.9% evanescent (rule 41, N=12)	CONFIRMED	Full enumeration

Dream/universe = ledger vs no-ledger; shape/value decoupling in dreams	PULL	Retrodicts phenomenology; channel structure
Scope = exhaust fraction	PULL	One variable, three regimes
Redundancy-objectivity convergence (quantum Darwinism)	PULL	Independent arrival; structural resonance only
Rendering mechanism physically instantiated (dream)	CONFIRMED	Biological existence proof, nightly
Waking universe as render	PULL	Mechanism exists; instance unproven
berlin52 exact optimum, raw, 20/20 genuine noise	CONFIRMED	Settled chain = optimal tour; unrounded 7544.37
Attractor path-independence at all N	CONFIRMED	Identical tours across seeds, on- and off-optimum
Attractor = global optimum in general	FALSIFIED	st70: 687 ≠ 675 after refinement
Fixed parameter values as Mold	DEAD	Collapse 1.00 → 1.10 across sweep
√N scaling law	PARTIAL	Rescues uniform (kroA100), fails structured (pr152)
Local resolution field K(x)	OPEN	Next carving in the rendering program
Hull order inside optimal tour	CONFIRMED	Verified live on berlin52
Inversion as replacement	FALSIFIED	Loses both compact instances
Dual read heads; handoff	CONFIRMED	Complementary wins; min-of-both ≤ 2.5% everywhere
Handoff law v2 (A* veto + density veto)	PULL	Blind 9/12, $p \approx 0.073$ — not significant; on record
Decisive-case pattern 7/8	POST HOC	Requires pre-registration, round 3
Arm convergence at ~3% plateau	PULL	New frontier; shared-channel exhaustion
Tension-fall (no-walker) solver	DEAD	= NN or worse (1.19–3.2); closure is load-bearing

Constraints select complexity	NOT SUPPORTED	Killed mean 0.186 > survivor 0.160; full census
BBP as random-access index; windings as exhaust	PULL	Base-specific; work-not-data correction applied
SM constants; $b_0=p^2$; gcd biconditional; gauge sequence	DEAD	Prior falsifications, retained on record

Part VIII — Open Nodes: The Frontier

Flux-form Law 3. Theorem 2 forces conservation into local flux; the corresponding filter — flips paid in displaced pairs — has not yet been imposed on the 34 survivors. It should thin them toward the few, and it is the next strike in the minimal substrate.

The 3% plateau. Both read heads exhaust one channel. Identifying the channel the residual lives in — or the third read head — is the frontier of the rendering program.

Round 3, margins pre-registered. Predict winner and decisiveness; score only what was declared decidable. The handoff law reaches significance or it dies there.

The local resolution field. K rendered from the field, not imposed on it: K(x) from local density, closing pr152's falsification from inside.

The ledger threshold. If measurement is first redundant encoding, the quantum/classical boundary is not a size threshold but a ledger threshold — the already-paid subset of the field. Redundancy is countable in principle. This is the framework's first pointed contact with the measurement seam.

What comes after, by the meta-theorem, cannot be listed. The frontier is open by construction; that opacity is what the frontier looks like from inside the cavity.

The universe begins as a gap with a mandate, pays for every difference into carriers that outlive the change, and reads its own receipts forward. The dream is what the same machine does when it stops paying. This document is one more paid mark.